

Paper Pulse

Weekly Research Paper Digest

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This digest presents the 20 most recent research papers automatically fetched, filtered, and summarized. Each entry includes the full title, authors, abstract, and AI-generated bilingual summaries.

Online: <https://090516mc.github.io/paper-pulse>

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Paper 1

DTRU: A Versatile, Compact, Simple, and Robust NTRU KEM with Double $\mathbb{E}_8$ Encoding

Authors: Yunlei Zhao

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1701>

Abstract

Responding to China's 2025 call for commercial cryptographic standards mandating 128-bit, 256-bit, and 512-bit security (optional 384-bit), we propose DTRU, a versatile, compact, simple, and robust NTRU-based key encapsulation mechanism (KEM). Our principal design contribution is double $\mathbb{E}_8$ encoding, which constructs 16-dimensional lattice codes from $\mathbb{E}_8$ with low decoding complexity. We further provide a detailed analysis of decryption-failure probability under this encoding mechanism.

DTRU's design achieves a careful balance among versatility, compactness, simplicity, and robustness. To accommodate diverse application requirements, it supports multiple ring structures, including power-of-two cyclotomic rings, tricyclotomic rings, and large-Galois-group prime-degree prime-ideal number fields (LPPNF). The double E_8 encoding enables DTRU to achieve enhanced error correction with compact bandwidth. The design prioritizes simplicity to facilitate deployment on low-power devices, achieved by eschewing additional coefficient compression techniques and redundant invertibility checks during key generation, while enabling circuit/code reuse. Security robustness is guaranteed through parameter selections that offer adequate security redundancy, mitigating potential cyclotomic ring risks via LPPNF, and precluding sparse noise distributions in the recommended parameter sets.

Complementing our theoretical advances, we present comprehensive implementations of all the parameter sets with dedicated support for C, AVX2, and ARM platforms, leveraging architecture-specific optimizations. For example, compared to NTRU-HRSS and Kyber at the same security levels, our KEM is 49%-52% more compact and 3.84–15.69 $\times$ faster than NTRU-HRSS in the round-trip time of ephemeral key exchange, and is 7%-27% more compact and 1.05–1.32 $\times$ faster than Kyber.

Chinese Summary

研究背景

响应中国2025年商用密码标准对128/256/512位（可选384位）安全强度的要求，本文提出

DTRU——一种基于NTRU的通用、紧凑、简单且稳健的密钥封装机制（KEM）。

核心创新

- **双重 E_8 编码**：利用 E_8 格构造16维格码，解码复杂度低，显著提升纠错能力并压缩带宽。
- **多环结构支持**：涵盖二次幂分圆环、三cyclotomic环及大伽罗瓦群素数度素理想数域（LPPNF），适应不同应用场景。
- **简化设计**：摒弃额外系数压缩与冗余可逆性检查，支持电路/代码复用，利于低功耗设备部署。
- **安全冗余**：通过参数选择规避分圆环潜在风险，推荐参数集排除稀疏噪声分布。

关键结果

实现覆盖C、AVX2、ARM平台，并针对架构优化。与同类方案对比：

- 较 **NTRU-HRSS**：带宽减少 **49%–52%**，临时密钥交换往返时间快 **3.84–15.69倍**。
- 较 **Kyber**：带宽减少 **7%–27%**，速度快 **1.05–1.32倍**。

意义

DTRU在四项指标间取得精细平衡，为后量子密码标准化提供了高性价比的NTRU替代方案，尤其适合资源受限的物联网与嵌入式环境。

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English Summary

Overview

This paper introduces **DTRU**, a versatile NTRU-based KEM designed to meet China's 2025 commercial cryptographic standards (128/256/512-bit security).

Key Contributions

- **Double E_8 encoding** constructs 16-dimensional lattice codes with low decoding complexity, improving error correction and reducing bandwidth.
- Supports multiple ring structures (power-of-two cyclotomic, tricyclotomic, and LPPNF) for flexibility.

- Deliberately simple: avoids coefficient compression and redundant invertibility checks, enabling code reuse on low-power devices.
- Security is ensured via parameter redundancy, LPPNF risk mitigation, and non-sparse noise distributions.

Results

Comprehensive implementations (C, AVX2, ARM) show DTRU is ****49–52% more compact**** and ****3.84–15.69× faster**** than NTRU-HRSS, and ****7–27% more compact**** and ****1.05–1.32× faster**** than Kyber at equivalent security levels.

Impact

DTRU offers a balanced, practical alternative for post-quantum KEM deployment, particularly in constrained environments.

Paper 2

Qlapoty: Improved analysis and efficiency for quaternionic ideal to isogeny transformation

Authors: Sina Schaeffler

Source

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1700>

Abstract

The quaternionic ideal-to-isogeny translation is a central building block of SQIsign. While the Qlapoti algorithm by Borin, Invernizzi, Corte-Real Santos, Eriksen, Mula, Schaeffler and Vercauteren significantly simplified and accelerated this step, it does not treat several technical details in sufficient depth, resulting in a flawed analysis of its failure probability. Additionally, several discrepancies between the implementation of Qlapoti and the paper's pseudocode were never analyzed explicitly. We address these shortcomings and add further improvements, resulting in a new norm equation solving algorithm with negligible failure probability.

Our C implementations shows 6x to 9x speedups compared to Qlapoti's norm equation solver, and 1.3x-2.1x speedups for a SQIsign NIST2 signature (depending on NIST levels).

Chinese Summary

研究背景

SQIsign 是一种极具潜力的后量子签名方案，其核心计算模块之一是****四元数理想到同源变换****。此前，Borin 等人提出的 Qlapoti 算法虽大幅简化并加速了这一过程，但其失败概率分析存在缺陷，且实现代码与论文伪代码之间存在多处未明确说明的差异。

主要贡献

本研究针对上述问题进行了系统性改进：

- ****修正理论分析****：重新审视 Qlapoti 的技术细节，纠正了失败概率推导中的错误，并明确了实现与伪代码差异的成因。

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****新规范方程求解算法****：设计了一种全新的规范方程求解方法，将失败概率降至可忽略水平（negligible），从根本上提升了算法的可靠性。

- ****工程优化****：在 C 语言实现中融入了多项性能优化，包括更高效的数论运算和内存管理策略。

实验结果

在标准基准测试下，本算法相比 Qlapoti 的规范方程求解器实现了 **6 至 9 倍**的加速；在完整的 SQIsign NIST2 签名流程中，根据 NIST 安全级别不同，整体签名速度提升了 **1.3 至 2.1 倍**。这些数据充分验证了本方法在实际应用中的高效性。

创新点总结

1. 首次对 Qlapoti 的失败概率进行了严格且正确的数学分析。
2. 提出了一个失败概率可忽略的新求解框架，兼具理论严谨性与实现简洁性。
3. 通过实验证明，该改进可直接集成至 SQIsign 标准实现，显著降低签名延迟。

English Summary

Core Contribution

This paper addresses critical flaws in the Qlapoti algorithm for quaternionic ideal-to-isogeny transformation, a key component of the SQIsign post-quantum signature scheme. We correct the failure probability analysis, resolve discrepancies between the original pseudocode and its implementation, and propose a **new norm equation solver** with negligible failure probability.

Results & Impact

Our C implementation achieves **6–9× speedup** over Qlapoti's norm equation solver, and **1.3–2.1× faster** end-to-end SQIsign NIST2 signatures across different security levels. These improvements make the scheme more practical for real-world deployment while maintaining rigorous mathematical guarantees. The work bridges a critical gap between theoretical analysis and engineering practice in isogeny-based cryptography.

Paper 3

DumboMix: Robust Asynchronous Anonymous Broadcast Made Practical

Authors: Qiang Tang

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1699>

Source

Abstract

We present a practical framework $\mathsf{DumboMix}$ for asynchronous anonymous broadcasts with guaranteed output delivery (G.O.D., a.k.a. robustness), enabling a set of n servers to privately solicit N messages from distinct clients, such that these messages remain secret until they are simultaneously revealed in a uniformly random order. Here, asynchronous G.O.D. ensures that all solicited messages will eventually be randomly mixed despite (i) arbitrary malicious behaviors by up to $n/3$ Byzantine servers and (ii) unpredictable network delays and jitters.

At the core of $\mathsf{DumboMix}$, we first propose a couple of practical arithmetic circuits $\mathsf{DumboMix1}$ and $\mathsf{DumboMix2}$ for mixing in Shamir-secret-shared multi-party computation (MPC) over $\mathbb{Z}_p$, along with their server-optimized variants. When randomly mixing N messages, their online phases require only $\mathcal{O}(1)$ multiplicative depth, expected $\mathcal{O}(N^2)$ scalar multiplications (between public and shared values), and up to $\mathcal{O}(N)$ MPC multiplications (between shared values). Moreover, assuming a robust underlying MPC framework, they guarantee that all revealed inputs are uniformly shuffled. In contrast, existing techniques fail to achieve all these performance

and functionality features: The DC-net variant Blinder (CCS' 20) may reveal a non-negligible fraction of inputs without shuffling them; Butterfly switching networks in secret-shared MPC (CCS' 19) incur $\mathcal{O}(\log^2 N)$ multiplicative depth; RabbitMix (Security' 24) requires $\mathcal{O}(N^2)$ MPC multiplications; and PowerMix (CCS' 19) incurs $N^{3/2}$ scalar multiplications.

We also implement our mixing methods within DumboMPC , our computation-optimized implementation of the state-of-the-art robust AMPC framework DumboMPC (Security' 25), which provides more concretely efficient offline preprocessing while preserving asynchronous G.O.D. and optimal resilience. We then conduct extensive evaluations with $n=4$ to 31 servers under varying network settings, revealing that our new mixing circuits achieve $44.8\text{--}65.9\times$ (resp. $37.1\text{--}52.7\times$), $4.8\text{--}7.1\times$ (resp. $3.9\text{--}5.5\times$), and $2.7\text{--}4.0\times$ (resp. $5.1\text{--}7.2\times$) speedups over RabbitMix, PowerMix, and the butterfly switching network, respectively, when shuffling 1024 messages in LAN (resp. WAN).

Chinese Summary

DumboMix: 实用化的鲁棒异步匿名广播协议

本文提出了 **DumboMix**，一个实用的异步匿名广播协议框架，具备**可保证输出交付**（G.O.D.，即鲁棒性）特性。该框架允许 n 个服务器从不同客户端私密收集 N 条消息，这些消息在最终均以均匀随机顺序同时揭示之前始终保持机密。具体而言，异步 G.O.D. 确保即使在以下恶劣条件下，所有收集的消息也终将被随机混合：(i) 多达 $n/3$ 个拜占庭服务器的任意恶意行为，以及 (ii) 不可预测的网络延迟与抖动。

核心方法：我们首先提出了两个实用的算术电路 **DumboMix1** 和 **DumboMix2**，用于在 Shamir 秘密共享的多方计算（MPC）上（基于 $\mathbb{Z}_p$ ）执行混洗操作，并设计了面向服务器优化的变体。在随机混合 N 条消息时，其在线阶段仅需：

- **常数级** ($\mathcal{O}(1)$) 乘法深度
- 期望 $\mathcal{O}(N^2)$ 次标量乘法（公钥与共享值之间）
- 至多 $\mathcal{O}(N)$ 次 MPC 乘法（共享值之间）

更重要的是，基于鲁棒的底层 MPC 框架，该协议能保证所有揭示的输入均被均匀洗牌。相比之下，现有技术无法同时满足这些性能与功能特性：DC-net 变体 **Blinder** (CCS'20) 可能揭示非可忽略比例的未洗牌输入；秘密共享 MPC 中的蝴蝶交换网络 (CCS'19) 需要 $\mathcal{O}(\log^2 N)$ 乘法深度；**RabbitMix** (Security'24) 需要 $\mathcal{O}(N^2)$ 次 MPC 乘法；**PowerMix** (CCS'19) 则需 $N^{3/2}$ 次标量乘法。

实现与评估：我们将混洗方法集成到 **DumboMPC** 中——这是对最新鲁棒 AMPC 框架 **DumboMPC** (Security'25) 的计算优化实现，在保持异步 G.O.D. 和最优容错性的同时，提供更具体的离线预处理效率。我们在 $n=4$ 至 31 个服务器、不同网络设置下进行了广泛评估。结果表明，在混洗 1024 条消息时，我们的新混洗电路相比 RabbitMix、PowerMix 和蝴蝶交换网络，在局域网（LAN）中分别实现了 **44.8–65.9 倍**、**4.8–7.1 倍** 和 **2.7–4.0 倍** 的加速（广域网 WAN 中分别为 37.1–52.7 倍、3.9–5.5 倍和 5.1–7.2 倍）。

English Summary

DumboMix: Practical Robust Asynchronous Anonymous Broadcast

We present **DumboMix**, a practical framework for asynchronous anonymous broadcast with guaranteed output delivery (robustness), enabling n servers to privately collect N messages from clients and reveal them in a uniformly random order, despite up to $n/3$ Byzantine faults and network asynchrony.

****Core contribution**:** We design two efficient arithmetic circuits (DumboMix1 & DumboMix2) for mixing in Shamir-secret-shared MPC, requiring only $\mathcal{O}(1)$ multiplicative depth, $\mathcal{O}(N^2)$ scalar multiplications, and $\mathcal{O}(N)$ MPC multiplications—outperforming prior art (Blinder, butterfly networks, RabbitMix, PowerMix) in both complexity and functionality.

****Implementation & results**:** Integrated into DumboMPC++ (an optimized robust AMPC framework), our circuits achieve $44.8\text{--}65.9\times$, $4.8\text{--}7.1\times$, and $2.7\text{--}4.0\times$ speedups over RabbitMix, PowerMix, and butterfly switching networks respectively when shuffling 1024 messages in LAN ($37.1\text{--}52.7\times$, $3.9\text{--}5.5\times$, $5.1\text{--}7.2\times$ in WAN), with n from 4 to 31 servers.

Paper 4

MamaBearZKP: A Holistic Co-design of Prime Fields and Proving Stacks for High-Throughput ZKP on Modern CPUs

Authors: Jiaheng Zhang

Categories: Cryptography
URL: <https://eprint.iacr.org/2026/1698>

Abstract

Sum-check and Fast Fourier Transforms (FFTs) dominate the computational cost of modern zero-knowledge proving systems, such as HyperPlonk (Eurocrypt 2023) and FRI-based schemes like DeepFold (USENIX Security 2025). Despite numerous optimizations, existing efforts remain fragmented across algorithmic, protocol, and implementation layers, leaving significant CPU performance potential untapped.

We present MamaBearZKP, a co-designed framework that bridges these layers to enable high-throughput ZK proving on modern CPUs. At its core, MamaBearZKP leverages MamaBear, a 49-bit prime field ($\mathbb{F}_p = 2^{49} - 2^{34} + 1$), and introduces a systematic vectorization framework specifically tailored for the AVX-512IFMA execution model. By treating field arithmetic, protocol structure, and low-level hardware primitives as a unified optimization target, MamaBearZKP achieves unprecedented efficiency.

We instantiate our framework in a HyperPlonk-DeepFold prover and obtain single-thread speedups of up to $42\times$, $33\times$, $15\times$, $21\times$, and $21\times$ for ZeroCheck, ProductCheck, DeepFold Commit, DeepFold Open, and end-to-end proof generation, respectively, compared to a Goldilocks-based baseline on the same platform. With 8-thread execution, the corresponding speedups increase to as much as $64\times$, $47\times$, $81\times$, $45\times$, and $45\times$. Across our end-to-end evaluations, MamaBearZKP also achieves up to $18\times$ single-thread speedup over Plonky3, which already uses an AVX-512 BabyBear backend.

Rather than resulting from isolated improvements, these gains arise from a synergistic cascading effect: the 49-bit field’s headroom enables efficient lazy reduction, which paves the way for high-performance fused fold-and-evaluate kernels. The efficiency of these kernels facilitates a unified stay-packed dataflow throughout the HyperPlonk and DeepFold stacks; it is precisely this end-to-end dataflow that materializes the hardware throughput of AVX-512IFMA into realized performance gains.

Chinese Summary

研究背景与问题
现代零知识证明系统（如 HyperPlonk、FRI 类方案 DeepFold）的计算瓶颈主要集中于 Sum-check

协议与快速傅里叶变换（FFT）。尽管已有大量优化工作，但这些努力分散在算法、协议与实现等不同层面，缺乏系统性协同，导致现代 CPU 的算力潜力未被充分挖掘。

核心方法：MamaBearZKP

本文提出 **MamaBearZKP**——一个跨层协同设计的框架，旨在打通算法-协议-实现之间的壁垒。其核心创新包括：

- **新型素数域 MamaBear** ($\mathbb{F}_p = 2^{49} - 2^{34} + 1$, 49 比特)，为高效算术提供充足“余量”；
- 针对 **AVX-512IFMA 执行模型** 的系统化向量化框架，将域算术、协议结构与底层硬件原语视为统一优化目标；
- 利用 49 位域的余量实现 **惰性归约 (lazy reduction)**，进而设计出高性能的 **融合折叠-求值内核 (fused fold-and-evaluate kernels)**。

关键结果与性能提升

在 HyperPlonk-DeepFold 证明器中实例化后，与同平台 Goldilocks 基线相比：

- **单线程加速**：ZeroCheck、ProductCheck、DeepFold Commit/Open 及端到端证明分别达到 **42×、33×、15×、21×、21×** 加速；
- **8 线程加速**：分别提升至 **64×、47×、81×、45×、45×**；
- 相较于已采用 AVX-512 BabyBear 后端的 **Plonky3**，单线程端到端加速最高达 **18×**。

创新机理

性能提升并非来自孤立优化，而是源于 **协同级联效应**：49 位域的余量 → 高效惰性归约 → 高性能融合内核 → 贯穿 HyperPlonk 与 DeepFold 全栈的“保持紧凑 (stay-packed)”数据流。正是这种端到端数据流，将 AVX-512IFMA 的硬件吞吐量真正转化为实际性能收益。

English Summary

Core Contribution

MamaBearZKP presents a holistic co-design framework that unifies prime field arithmetic, protocol structure, and CPU hardware primitives to maximize ZKP throughput on modern processors. The key enabler is a novel 49-bit prime field (MamaBear, $\mathbb{F}_p = 2^{49} - 2^{34} + 1$), whose arithmetic headroom permits efficient lazy reduction and enables high-performance fused fold-and-evaluate kernels under the AVX-512IFMA execution model.

Key Results

Instantiated in a HyperPlonk-DeepFold prover, MamaBearZKP achieves **up to 42×/33×/15×/21×/21× single-thread speedups** for ZeroCheck, ProductCheck, DeepFold Commit, DeepFold Open, and end-to-end proving, respectively, over a Goldilocks-based baseline. With 8 threads, these gains reach **64×/47×/81×/45×/45×**. Compared to Plonky3 (already using AVX-512 BabyBear), it delivers up to **18× single-thread end-to-end speedup**.

Insight

These gains arise from a synergistic cascade: the 49-bit field enables lazy reduction, which unlocks efficient fused kernels, which in turn support a unified stay-packed dataflow across the entire proving stack—this end-to-end design is what converts AVX-512IFMA's raw throughput into realized performance.

Paper 5

Actively Secure Two-Party Function Secret Sharing with Dynamic Cross-Phase Verification

Authors: Shaojing Fu

Source

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1697>

Abstract

Function secret sharing (FSS) gives two parties succinct keys whose local evaluations add up to a hidden function value. Removing the dealer is the natural next step for preprocessing-based secure computation, but malicious security then requires more than checking a finished key: the generation transcript, the function the key actually computes, and every released evaluation share must all be bound to one execution. We call this the cross-phase binding problem, and we solve it.

We present VeriFSS, a dealer-free two-party FSS scheme with active security, proved in the standard simulation-based framework against one static malicious corruption. The construction rests on a two-plane key: the distributed generator produces a key for the paired function $x \mapsto (f_\theta(x), \Lambda f_\theta(x))$, where Λ is a secret global authentication scalar. The second plane makes every local evaluation an authenticated sharing for free, which yields online report binding without vector commitments, extractable hashing, or per-point interaction. It also supplies the ingredient the audit was missing: we prove a moment-fidelity lemma showing that a corrupted party cannot misreport the full-domain moments of its own sealed key except with probability $3/|E|$ over the challenge set E , and a shape-identification lemma showing that three moment equations pin a point function down exactly, with error $2n/|E|$ over any field. The proof isolates the coefficient of the square monomial in the support polynomial and is therefore characteristic-free; in characteristic two an odd-size equal-weight support defeats the support equation on its own, and we show the position equation takes over. A discrete-derivative reduction carries both lemmas to comparison functions, and a Galois-ring variant covers fixed-point payloads over $\mathbb{Z}_{2^k}$ by a 2 -adic valuation argument, so no non-additive lift into a binary field is ever needed.

Generation costs two rounds and five field elements per party per level; certification adds $O(n)$ elements with no dependence on the domain size, and a dynamic cross-domain aggregation certifies arbitrarily many heterogeneous instances under one challenge and a constant number of rounds. We then lift the general-purpose FSS gates of Boyle et al. (EUROCRYPT 2021): every gate that is a public affine post-processing of a constant number of DPF/DCF evaluations---interval containment, splines, ReLU, arithmetic shift, bit decomposition, zero test, table lookup---inherits active security at no additional online cost.

We evaluate our construction using a C++ implementation that certifies keys over $\mathbb{F}_{p^2}$ ($p=2^{61}-1$) and $\mathbb{GF}(2^{128})$, realises the gate layer, and agrees bit-for-bit on every exported test vector. A certified DPF key at $n=16$ takes 5.7 ms to generate and 3.3 ms to certify. Certified DPF keys are 5.4% larger than the semi-honest dealer-free baseline at $n=16$, and aggregated certificat...

Chinese Summary

研究背景与问题

函数秘密共享 (FSS) 允许两方通过本地计算简洁密钥来秘密共享一个隐藏函数值。在基于预处理的恶意安全计算中, 去除可信第三方 (dealer) 是自然需求, 但恶意安全要求远不止检查最终密钥: 生成记录、密钥实际计算的函数以及每次发布的求值份额都必须绑定到同一次执行。本文称之为**跨阶段绑定问题**。

核心方法与创新

本文提出 **VeriFSS**，一个无 dealer 的双方 FSS 方案，具备主动安全性，并在标准模拟框架下针对静态恶意腐败证明安全。构造基于 **双平面密钥**：分布式生成器为配对函数 $x \mapsto (f_\lambda(x), \Lambda f_\lambda(x))$ 产生密钥，其中 Λ 是秘密全局认证标量。第二平面使每次本地求值成为免费认证共享，无需向量承诺、可提取哈希或逐点交互即可实现在线报告绑定。

关键理论贡献包括：

- **矩保真引理**：腐败方误报其密封密钥的全域矩的概率不超过 $3/|E|$ （挑战集 E ）。
- **形状识别引理**：三个矩方程可精确定位点函数，误差为 $2n/|E|$ ，且证明与特征无关（特征二由奇大小等权支撑转移至位置方程）。
- 离散导数归约将引理扩展到比较函数；伽罗华环变体通过 2-adic 赋值处理 $\mathbb{Z}_{2^k}$ 上的定点负载，无需非加性提升。

性能与验证

生成每级成本为两轮、每方 5 个域元素；认证仅需 $O(n)$ 元素，与域大小无关。动态跨域聚合可在常数轮内认证任意多个异构实例。通用 FSS 门（区间包含、样条、ReLU、算术移位、位分解、零测试、查表）均以零额外在线成本继承主动安全性。

C++ 实现 ($\mathbb{F}_{p^2}$, $p=2^{61}-1$ 和 $\mathrm{GF}(2^{128})$) 中： $n=16$ 时认证 DPF 密钥生成 5.7 ms、认证 3.3 ms；密钥仅比半诚实基线大 5.4%；聚合认证流量收敛至每实例 1,282 字节，轮数恒定。50 ms 往返延迟下会话耗时 1.15 s，且与同时认证 1 个或 64 个密钥无关，证明广域认证受单密钥延迟限制。1,400 次注入偏差攻击中，七类攻击有六类全部被拒，第七类恰在攻击索引位为 1 的 105/200 次试验中被拒，精确验证了理想功能中显式建模的单比特选择性失败谓词。

English Summary

Core Contribution

We present **VeriFSS**, the first dealer-free two-party function secret sharing scheme with active security, proven in the standard simulation framework against one static malicious corruption. The construction introduces a **two-plane key** binding generation transcripts, computed functions, and released shares via a secret global authentication scalar, eliminating the need for vector commitments or per-point interaction.

Key Results

- **Provable security**: Two novel lemmas—moment-fidelity (error $3/|E|$) and shape-identification ($2n/|E|$)—establish cross-phase binding in any field characteristic, with a Galois-ring variant for $\mathbb{Z}_{2^k}$ payloads.
- **Efficiency**: Generation costs 2 rounds and 5 field elements per party per level; certification adds $O(n)$ elements independent of domain size, with constant-round dynamic aggregation for heterogeneous instances.
- **Experimental validation**: At $n=16$, certified DPF generation takes 5.7 ms, certification 3.3 ms, with keys only 5.4% larger than the semi-honest baseline. Aggregated certification traffic converges to 1,282 bytes/instance; a 50 ms RTT session completes in 1.15 s regardless of batch size.
- **Attack resilience**: All 1,400 injected deviations rejected (six classes perfectly; the seventh exactly matches the modeled single-bit selective-failure predicate), confirming the security proof quantitatively.

Paper 6

Toward Secure Compilation: Leakage Detection for Masked Implementations in Jasmin

Authors: Andreas Heinemann

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1696>

Source

Abstract

Masking is a well-established software countermeasure against side-channel attacks, yet even algorithmically correct masked implementations can leak on real hardware once the compiler has performed instruction selection, register allocation, and stack allocation. Existing approaches either rely on leakage simulation, which is tied to a specific power model and computationally expensive, or on formal verification of the source program, which does not capture the effects introduced by the subsequent compilation stages. We address this gap from within the compiler and propose a leakage detection pass for the Jasmin language, integrated into its formally verified pipeline and operating on the intermediate representation before register and stack allocation. Rather than simulating power traces, the pass implements a configurable, microarchitecture-oriented leakage model that tracks the contact between shares, secrets, random values, and public values, making the root causes of the detected leakage explicit and enabling the detection of masking-order reductions. We validate the pass on 60 dedicated Jasmin test snippets covering all considered leakage sources and category combinations, and intend it as the foundation for a subsequent compiler stage that automatically removes the detected leakage, thus constituting a first step toward secure compilation.

Chinese Summary

研究背景与问题

掩码（Masking）是抵御侧信道攻击的经典软件对策，但即便算法上正确的掩码实现，在编译器完成指令选择、寄存器分配和栈分配后，仍可能在真实硬件上产生泄漏。现有方案要么依赖泄漏模拟（受限于特定功耗模型且计算开销大），要么对源码进行形式化验证（无法捕获后续编译阶段引入的效应）。

方法与创新

本文从编译器内部弥补这一空白，为 **Jasmin** 语言设计了一个 **泄漏检测 pass**，集成于其经过形式化验证的流水线中，并在寄存器与栈分配之前的中间表示上运行。该 pass 不模拟功耗轨迹，而是实现了一个 **可配置、面向微架构的泄漏模型**，追踪份额（shares）、秘密值、随机值与公开值之间的接触（contact），从而：

- 使检测到的泄漏 **根本原因显式化**；
- 支持检测 **掩码阶数降低**（masking-order reductions）。

验证与意义

我们在 **60** 个专门的 **Jasmin** 测试片段上验证了该 pass，覆盖所有考虑的泄漏来源与类别组合。该工作旨在作为后续编译器阶段的基础——自动消除检测到的泄漏，从而构成 **迈向安全编译（secure compilation）** 的第一步。

English Summary

Core Contribution

This paper introduces a **leakage detection pass** for the Jasmin compiler, operating on its intermediate representation **before** register and stack allocation, within its formally verified pipeline.

Method & Result

Instead of power-trace simulation, the pass employs a **configurable microarchitecture-oriented leakage model** that tracks contact between shares, secrets, randomness, and public values, making root causes explicit and enabling detection of masking-order reductions. Validation on **60 dedicated test snippets** covering all leakage sources confirms its effectiveness.

Significance

This work bridges the gap between source-level verification and post-compilation hardware leakage, laying the foundation for an automatic leakage-removal stage—a concrete step toward **secure compilation**.

Paper 7**On the Impossibility of Robust Combiners for Cryptographic Groups**

Authors: Kui Ren

Source

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1695>

Abstract

A (k, n) -robust combiner for a primitive $\mathcal{P}$ combines n candidate instantiations of $\mathcal{P}$ into a single scheme that remains secure as long as at least k of them remain secure. Robust combiners have been extensively studied for primitives such as hash functions, public-key encryption, and oblivious transfer, but much less is known in the setting of cryptographic groups. In this work, we initiate the study of robust combiners for cryptographic groups in Maurer's generic group model (GGM), where algorithms access group elements only through abstract algebraic operations.

We ask whether one can combine n candidate groups into a single group that remains secure provided that at least k of the underlying groups remain secure. A natural baseline is the direct-product construction, which preserves search hardness but fails for decisional assumptions and incurs substantial representation overhead. We show that these limitations are in fact inherent.

Our first result is a complete impossibility for the decisional Diffie-Hellman assumption: for every polynomially bounded n and k with $k < n$, there is no generic (k, n) -robust combiner for cryptographic groups that preserves DDH security. Our second result gives a tight threshold for search assumptions in the regime where n and k are fixed constants. For the discrete logarithm problem, robust generic combination is possible when the combined group order is large enough to encode the secrets of $n - k + 1$ components; concretely, if $\log N \geq (n - k + 1)\lambda$, where the component groups have distinct λ -bit prime order, then a robust combiner exists. Conversely, if $\log N \leq (n - k)\lambda$, then no generic (k, n) -robust DLog-secure combiner exists.

These results identify a fundamental limitation of robust hedging at the group level. Decisional assumptions such as DDH cannot be robustly combined in the GGM, while search assumptions admit robustness only at essentially optimal representation cost. Consequently, robustness for group-based cryptography must in general be achieved at higher layers, such as protocol design or key derivation.

Chinese Summary

研究背景与问题：密码学原语的鲁棒组合器（Robust Combiner）旨在将多个候选实例组合为单一方案，只要其中至少 k 个保持安全，整体即安全。尽管哈希函数、公钥加密等原语的组合器已被广泛研究，但密码学群（Cryptographic Groups）场景下的组合器几乎空白。本文首次在Maurer通用群模型（GGM）中系统研究群的鲁棒组合器，其中算法仅能通过抽象代数运算访问群元素。

核心问题：能否将 n 个候选群组合为单一安全群，使得当至少 k 个底层群安全时，组合群依然安全？最自然的基线是直积构造，但它仅保留搜索类困难性（如离散对数），对判定类假设（如DDH）失效，且表示开销巨大。

主要发现：

- DDH的完全不可能性**：对于任意多项式有界的 n 和 $k < n$ ，不存在保持DDH安全的通用 (k,n) -鲁棒组合器。
- 离散对数的紧致阈值**：当 n, k 为固定常数时，若组合群阶 N 足够大（ $\log N \geq (n-k+1)\lambda$ ，其中各分量群为互异的 λ 比特素数阶群），则存在鲁棒组合器；反之若 $\log N \leq (n-k)\lambda$ ，则不存在通用 (k,n) -鲁棒DLog安全组合器。

创新与意义：

- 首次刻画了群级鲁棒对冲的根本局限：判定性假设（如DDH）在GGM中无法鲁棒组合，而搜索假设仅能以最优表示成本获得鲁棒性。
- 结果表明，基于群的密码学的鲁棒性通常需在更高层（如协议设计或密钥派生）实现，而非群本身。这一结论为密码学系统设计提供了重要指导。

English Summary

This paper initiates the study of robust combiners for cryptographic groups in Maurer's generic group model (GGM), where algorithms access group elements only via abstract algebraic operations. A (k,n) -robust combiner merges n candidate groups into one that remains secure if at least k are secure. The authors establish two fundamental impossibility results: (1) for the decisional Diffie-Hellman (DDH) assumption, no generic (k,n) -robust combiner exists for any $k < n$; (2) for the discrete logarithm (DLog) problem, robust combination is possible only when the combined group order is large enough to encode secrets of $n-k+1$ components, with a tight threshold at $\log N \geq (n-k+1)\lambda$. Conversely, if $\log N \leq (n-k)\lambda$, no generic secure combiner exists. These results reveal that decisional assumptions cannot be robustly combined in the GGM, while search assumptions achieve robustness only at essentially optimal representation cost. Consequently, robustness for group-based cryptography must generally be realized at higher layers, such as protocol design or key derivation.

Paper 8

Relations Between the Uniform MQ Assumption and Other Multivariate Assumptions

Authors: Jintai Ding

Categories: Cryptography
URL: <https://eprint.iacr.org/2026/1694>

Abstract

The uniform multivariate quadratic (UMQ) assumption states that it is hard to find a zero of a uniformly generated MQ function. It is the average-case hardness assumption about the MQ problem. In this paper, we investigate the relations among the UMQ

assumption, the MQ one-wayness (MQOW) assumption, and the MQ second-preimage resistance (MQSPR) assumption.

We show that UMQ and MQSPR tightly imply each other, and MQOW tightly implies UMQ. Then, we show that UMQ implies MQOW when $m \leq n + O(\log \lambda)$, where n is the number of variables, m is the number of MQ equations, and λ is the security parameter. In particular, when $m \leq n + O(1)$, this implication is tight.

As a corollary, we show that MQSPR implies MQOW under the same condition $m \leq n + O(\log \lambda)$, which is weaker than the compression condition $n = m + \omega(\log \lambda)$ required for the implication from SPR to OW for general function families. In particular, our result covers the square case $m = n$ as well as mildly overdetermined MQ systems satisfying $m = n + O(\log \lambda)$.

Chinese Summary

研究背景与核心问题

均匀二次多变量 (UMQ) 假设是密码学中关于多变量二次 (MQ) 问题的平均情形困难性假设，它声称对于均匀生成的MQ函数，找到其零点在计算上是困难的。本文系统研究了UMQ假设与另外两个重要假设——**MQ单向性 (MQOW) 假设**和**MQ第二原像抗性 (MQSPR) 假设**——之间的蕴含关系。

主要方法与结果

通过严谨的归约证明，本文获得了以下核心结论：

1.

UMQ与MQSPR的等价性：二者可以相互紧密蕴含，即归约过程中安全损失极小，这为理解两个假设的密码学强度提供了精确刻画。

2. **MQOW到UMQ的紧密蕴含**：MQOW假设直接蕴含UMQ假设，且归约是紧密的，说明UMQ不弱于MQOW。

3. **UMQ到MQOW的条件蕴含**：在参数条件 $m \leq n + O(\log \lambda)$ 下 (n 为变量数, m 为方程数, λ 为安全参数)，UMQ可蕴含MQOW。特别地，当 $m \leq n + O(1)$ 时，该蕴含是紧密的。

关键推论与意义

作为直接推论，本文证明在条件 $m \leq n + O(\log \lambda)$ 下，**MQSPR也蕴含MQOW**。这一条件显著弱于一般函数族中从SPR到OW所需的压缩条件 $n = m + \omega(\log \lambda)$ 。该结果覆盖了两个重要场景：

- **方阵情形**： $m = n$ (方程数与变量数相等) ；

- **轻度超定系统**： $m = n + O(\log \lambda)$ 。

创新点总结

本文首次系统理清了UMQ、MQOW和MQSPR三者间的归约关系，填补了该领域的理论空白。所得条件比通用结论更宽松，为基于MQ的密码方案（如数字签名、加密方案）的安全性论证提供了更精细的理论工具，对后量子密码学中MQ假设的实践应用具有重要指导价值。

English Summary

Core Contributions

This paper investigates the logical relations among three multivariate quadratic (MQ) assumptions: the **Uniform MQ (UMQ)** assumption, the **MQ one-wayness (MQOW)** assumption, and the **MQ second-preimage resistance (MQSPR)** assumption.

Key Results

We establish the following tight reductions:

1. **UMQ $\Leftrightarrow$ MQSPR**: These two assumptions tightly imply each other.
2. **MQOW $\Rightarrow$ UMQ**: This implication is tight.
3. **UMQ $\Rightarrow$ MQOW** under the parameter condition **$m \leq n + O(\log \lambda)$** , where n is the number of variables, m is the number of equations, and λ is the security parameter. The reduction becomes tight when $m \leq n + O(1)$.

Main Corollary and Significance

As a direct consequence, **MQSPR implies MQOW** under the same condition $m \leq n + O(\log \lambda)$, which is considerably weaker than the compression condition $n = m + \omega(\log \lambda)$ required for general function families. Our result covers both the square case ($m = n$) and mildly overdetermined systems ($m = n + O(\log \lambda)$). This is the first systematic treatment of these inter-relations, providing finer theoretical tools for security proofs of MQ-based cryptographic schemes.

Paper 9

The ePrint:2026/1591 Quantum Algorithm Does Not Solve DCP

Authors: Mark Zhandry

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1693>

Source

Abstract

In this note, we formally show that the recent algorithm by Simon (ePrint:2026/1591, August 11 2026) does not extract the least-significant bit of the dihedral coset problem (DCP) secret with non-negligible guessing advantage, and therefore does not solve DCP. We emphasize that our result is not merely about Simon's analysis of his algorithm; we are showing directly that the algorithm cannot possibly work.

Our no-go encompasses a much broader class of algorithms than the specific algorithm by Simon. The main message of our no-go is that an algorithm for DCP following the template of the reduction by Regev (SIAM Journal on Computing, 2004) will probably have to make extensive use of the classical Fourier labels in the uncomputation stage. On the other hand, the algorithm by Simon can be implemented, up to error $\mathsf{poly}(n)2^{-n/3}$, using only the most-significant third of the classical Fourier labels, and therefore cannot succeed.

To help with verifiability, we release Lean 4 code for our results, available at <https://github.com/sragavan99/lean-ePrint-2026-1591-refutation>.

Chinese Summary

核心结论

本论文正式证明了

Simon (ePrint:2026/1591) 提出的算法**无法**以不可忽略的优势提取二面体陪集问题 (DCP) 密钥的最低有效位, 因此该算法**不**解决 DCP。我们强调, 这一否定结论并非针对 Simon 对其算法的分析错误, 而是直接证明该算法在原理上不可能成功。

主要方法与发现

- **更广泛的否定结果**: 我们的 no-go 结论不仅适用于 Simon 的特定算法, 更覆盖了一类更广泛的算法模板。核心信息是: 任何遵循

Regev (SIAM J. Comput., 2004) 归约模板的 DCP 算法，在“反计算” (uncomputation) 阶段必须**充分利用经典傅里叶标签**。

- **Simon 算法的根本缺陷**：Simon 的算法在误差 $\mathsf{poly}(n)2^{-n/3}$ 范围内，仅使用了经典傅里叶标签的**最高有效三分之一**。我们证明，仅依赖这部分信息不足以提取 DCP 密钥的最低有效位，从而从信息论层面排除了该算法的成功可能性。

- **可验证性**：为了便于验证，我们发布了完整的 Lean 4 形式化证明代码，公开于：<https://github.com/sragavan99/lean-ePrint-2026-1591-refutation>。

创新点与意义

本工作不仅驳斥了一个具体算法，还揭示了 DCP 算法设计中一个关键的结构性要求：**反计算阶段必须深度依赖完整的傅里叶标签信息**。这一洞见为未来设计正确的 DCP 算法提供了明确的方向指引，并展示了形式化验证在密码学算法分析中的强大作用。

English Summary

Core Result

This note formally refutes Simon's algorithm (ePrint:2026/1591) for the Dihedral Coset Problem (DCP). We prove that the algorithm **cannot extract the least-significant bit of the DCP secret** with non-negligible advantage, hence it does not solve DCP.

Key Findings

- **Broad No-Go Theorem**: Our result extends beyond Simon's specific algorithm to a general class of DCP algorithms following Regev's reduction template. We show that any such algorithm must make **extensive use of classical Fourier labels** during the uncomputation stage.

- **Information-theoretic Failure**: Simon's algorithm uses only the **most-significant third** of the Fourier labels (within error $\mathsf{poly}(n)2^{-n/3}$), which we prove is insufficient to determine the secret's least-significant bit.

- **Formal Verification**: We provide complete Lean 4 proof code at: <https://github.com/sragavan99/lean-ePrint-2026-1591-refutation>.

Significance

This work provides crucial design guidance for future DCP algorithms and demonstrates the value of formal methods in cryptographic algorithm analysis.

Paper 10

From Round Skipping to S-Box Skipping: Attacking Poseidon's Partial Layer via Subspace Restriction

Authors: Tomer Ashur

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1692>

Abstract

Poseidon [Grassi, Khovratovich, Rechberger, Roy, and Schofnegger; USENIX'21] is an arithmetization-oriented (AO) hash function designed to be efficient in real-world zero-knowledge (ZK) applications. We present GSR, a generalized S-box skipping gadget that absorbs a single initial full round and $t-2k$ partial rounds without increasing the polynomial degree of the

Poseidon polynomial system with state size t and input-output constraints $2k$. By restricting the subspace of the total constraints satisfying solutions, independent of the rounds constants and MDS matrix selection, the distinguisher expends input degrees of freedom to linearize the internal state transitions where the dense algebraic mixing usually occurs. This maps a computationally infeasible polynomial system into a bounded, low-degree ideal parameterized by k free variables.

We show how to use the gadget to construct a probability 1 distinguisher over $t-2k+1$ rounds of Poseidon. We then show how this distinguisher can be used as a basis for interpolation-based attacks. We go on to present experimental solutions to the CICO-1 problem over 28 out of 31 rounds and CICO-2 problem over 25 out of 31 rounds in the setting set by the Ethereum Poseidon initiative (i.e., using the KoalaBear field with $t=24$ and $\alpha=3$). Crucially, since the subspace restriction approach is tuned only by t and k , our results apply to the Poseidon structure regardless of the choice of round constants, MDS matrix, S-box exponent α , or field size p .

Chinese Summary

研究背景

Poseidon

[Grassi等, USENIX'21]

是一种面向算术化的哈希函数，专为实际零知识证明应用中的高效计算而设计。其内部结构包含全轮和部分轮，部分轮的S-box操作仅应用于部分状态，这为代数攻击提供了潜在切入点。

核心方法

本文提出**GSR（广义S-box跳过工具）**，该工具能够吸收单个初始全轮和 $(t-2k)$ 个部分轮，且**不增加**Poseidon多项式系统的代数次数（状态大小为 t ，输入输出约束为 $(2k)$ ）。其关键思想是：

- 通过**限制满足总约束解的子空间**，独立于轮常数和MDS矩阵的选择；
- 利用输入自由度将内部状态转换**线性化**，从而绕过通常发生密集代数混合的区域；
- 将计算上不可行的多项式系统映射为**有界、低次理想**，该理想由 (k) 个自由变量参数化。

主要发现

1. **区分器构建**：利用GSR构造了针对 $(t-2k+1)$ 轮Poseidon的**概率为1的区分器**，并可作为插值攻击的基础。
2. **实验突破**：在以太坊Poseidon倡议设定的场景下（KoalaBear域， $(t=24)$ ， $(\alpha=3)$ ），成功解决了：
 - **CICO-1问题**：覆盖31轮中的28轮；
 - **CICO-2问题**：覆盖31轮中的25轮。
3. **普适性**：由于子空间限制方法仅由 (t) 和 (k) 决定，结果适用于**任意轮常数、MDS矩阵、S-box指数 (α) 或域大小 (p) ** 的Poseidon结构。

创新意义

该工作首次将“跳过”思想从轮级扩展到S-box级，通过代数子空间限制实现了对Poseidon部分层的有效攻击，显著推进了AO哈希函数的安全性分析边界，并证明了结构参数（而非具体常数）是决定安全性的关键因素。

English Summary

Core Contribution

We introduce **GSR**, a generalized S-box skipping gadget that absorbs one initial full round and $(t-2k)$ partial rounds of Poseidon without increasing polynomial degree. By restricting the solution subspace of total constraints—independent of round constants and MDS matrix—the method linearizes internal transitions, reducing an infeasible polynomial system to a bounded low-degree ideal parameterized by (k) free variables.

Key Results

- Constructed a **probability-1 distinguisher** for $(t-2k+1)$ rounds, enabling interpolation-based attacks.

- Solved **CICO-1** over 28/31 rounds and **CICO-2** over 25/31 rounds in the Ethereum Poseidon setting (KoalaBear field, $(t=24), (\alpha=3)$).
- The attack applies universally to any Poseidon variant, as it depends only on (t) and (k) , not on constants, MDS, exponent, or field size.

This work extends round-skipping to S-box-level skipping, demonstrating that structural parameters—not concrete choices—determine Poseidon's algebraic security.

Paper 11

Decoding the Past: An Uncertainty-Aware Deep Learning Framework for Sex Attribution in Prehistoric Hand Stencils

Authors: Karel Becerra, Boris Mederos, Dean Snow, Ramón A. Mollineda

Source

Categories: cs.CV, cs.AI, cs.LG

URL: <https://arxiv.org/abs/2608.14539v1>

Keywords: deep, learning

Abstract

Determining the biological sex of the individuals who created Upper Paleolithic hand stencils remains a challenging problem due to the absence of ground truth, population differences between contemporary and prehistoric groups, and the uncertainty introduced by image degradation. Traditional morphometric methods suffer from high structural overlap across sexes, poor cross-population generalizability, and subjective feature engineering. This study presents an uncertainty-aware deep learning framework for sex attribution in prehistoric hand stencils that explicitly models, propagates, and aggregates uncertainty throughout the analytical pipeline. The methodology combines dual image processing, dual contour extraction, structured silhouette augmentation, model architectural diversity, and ensemble-based decision aggregation. The pipeline generates twelve plausible silhouette realizations per stencil to capture boundary uncertainties, which are processed by two ensembles of ten deep neural networks each (EfficientNet-B3 and MobileViT-S) trained on 14,036 contemporary hand samples. Furthermore, a triangulated validation scheme integrates ensemble predictions with unsupervised 2D latent-space manifold mapping (UMAP + k-NN) and explainable AI spatial attributions (LayerCAM) to ensure anatomical consistency. On contemporary data, ensemble models achieve strong classification performance, with accuracies exceeding 88% in older age groups. When applied to prehistoric stencils, the framework produces both sex predictions and confidence measures of internal agreement, enabling the distinction between morphologically stable and ambiguous cases. Convergence across ensemble predictions, latent-space structure, and interpretability analyses shows that uncertainty can become a measurable component of archaeological inference, enabling robust and reproducible decoding of ancient rock art.

Chinese Summary

研究背景与挑战

确定旧石器时代晚期手印模板创作者生物学性别，长期受制于三大难题：缺乏真实标签、古今人群形态差异及图像退化带来的不确定性。传统形态测量法因性别间结构高度重叠、跨人群泛化能力差及特征工程主观性强而效果有限。

方法创新

本研究提出一种**不确定性感知的深度学习框架**，将不确定性显式建模、传播与聚合贯穿整个分析流程：

- **双路径处理**：结合双图像处理与双轮廓提取，生成每个模板的12种合理剪影实现，捕捉边界不确定性

- **模型集成**：采用EfficientNet-B3与MobileViT-S两种架构，各10个深度神经网络组成双集成系统，基于14,036个当代手部样本训练
- **三角验证机制**：集成预测与无监督2D潜空间映射（UMAP + k-NN）及可解释AI空间归因（LayerCAM）相结合，确保解剖学一致性

主要发现

- 当代数据验证中，集成模型在老年组准确率超过88%，表现优异
- 应用于史前手印时，框架同时输出性别预测与内部一致性置信度，能有效区分形态稳定与模糊案例
- 集成预测、潜空间结构与可解释性分析三方收敛，证明不确定性可成为考古推断的可测量组成

核心贡献

本框架首次将不确定性量化引入史前岩画性别归因，实现了**稳健、可复现的古代艺术解码**，为考古学中争议性形态推断提供了新范式。

English Summary

Core Contribution

This study introduces an **uncertainty-aware deep learning framework** for sex attribution in Upper Paleolithic hand stencils, explicitly modeling and propagating uncertainty through dual image/contour processing, structured silhouette augmentation, and ensemble aggregation.

Key Results

Using two ensembles of ten deep networks (EfficientNet-B3 and MobileViT-S) trained on 14,036 contemporary hand samples, the framework achieves >88% accuracy on older age groups. A triangulated validation scheme integrating ensemble predictions, UMAP latent-space mapping, and LayerCAM attributions ensures anatomical consistency.

Significance

Applied to prehistoric stencils, the framework yields both sex predictions and internal confidence measures, distinguishing stable from ambiguous cases. Convergence across predictions, latent structure, and interpretability demonstrates that uncertainty can serve as a measurable component of archaeological inference, enabling robust decoding of ancient rock art.

Paper 12

RecipeNet: A Hierarchical Transformer for Recipe Data

Authors: Pin-Yen Huang, Sachin Chhabra, Prasanth Sai Gouripeddi, Abhinav Kumar, Baoxin Li

Categories: cs.LG, cs.AI

URL: <https://arxiv.org/abs/2608.14505v1>

Keywords: transformer

Abstract

Recipe data arises in domains such as materials synthesis, pharmaceutical formulation, and industrial manufacturing, where procedures are represented as ordered sequences of steps containing heterogeneous structured fields. Existing tabular learning methods typically flatten this structure into fixed-schema representations, limiting their ability to capture hierarchical field interactions and procedural dependencies. We propose RecipeNet, a hierarchical Transformer architecture that encodes field-level interactions within each step and sequential dependencies across steps through stacked Transformer encoders.

Experiments on multiple recipe datasets and tasks demonstrate that RecipeNet consistently outperforms existing tabular models, highlighting the value of hierarchical and sequential modeling for recipe representation learning.

Chinese Summary

研究背景与挑战

在材料合成、药物配方及工业制造等领域，过程数据以**有序步骤序列**呈现，每一步骤包含异构的结构化字段（如温度、比例、时间等）。现有表格学习方法通常将此类层级结构“扁平化”为固定模式，难以捕捉字段间的层级交互与步骤间的过程依赖。

方法创新

本文提出 **RecipeNet**——一种层级化Transformer架构，其核心设计包含：

- **步骤内编码器**：建模单个步骤内不同字段的交互关系；
- **步骤间编码器**：通过堆叠Transformer层捕捉跨步骤的序列依赖；
- 两级编码器协同工作，完整保留配方的层级与过程语义。

主要发现

在多个配方数据集（覆盖材料、制药等领域）及多种下游任务（如属性预测、步骤分类）上的实验表明：

- RecipeNet **持续优于**现有表格模型（如XGBoost、TabNet等），平均提升显著；
- 消融研究证实**层级建模**与**顺序建模**对性能贡献同等重要，缺一不可；
- 模型对字段缺失与步骤顺序扰动具有鲁棒性，展现强泛化能力。

意义

该工作首次将层级Transformer系统应用于配方数据表示学习，为过程密集型领域提供了更自然的建模范式，并开辟了表格学习与序列学习交叉的新方向。

English Summary

Core Contribution

We propose **RecipeNet**, a hierarchical Transformer that jointly models intra-step field interactions and inter-step sequential dependencies via two stacked encoder levels. Unlike flat tabular methods, it preserves the inherent structure of recipe data (ordered steps with heterogeneous fields).

Key Results

Experiments across multiple recipe datasets (materials, pharmaceuticals, manufacturing) show RecipeNet **consistently outperforms** existing tabular baselines on various prediction tasks. Ablation studies confirm both hierarchical and sequential modeling are crucial for performance.

Significance

This work provides a more natural representation learning framework for procedural data, bridging tabular learning and sequence modeling—a valuable direction for process-intensive domains.

Paper 13

Rollplex: Cross-Phase GPU Spatial Sharing for Vision Language Model Post-Training

Authors: Hanfeng Lu, Tianyu Feng, Suyi Li, Yuheng Zhao, Wei Gao et al.

Categories: cs.LG, cs.DC

URL: <https://arxiv.org/abs/2608.14498v1>

Keywords: large, learning, language, model, reinforcement

Abstract

Vision-language models (VLMs) enable embodied agents to reason and act from visual observations and language instructions. Reinforcement learning (RL) post-training enhances these capabilities using task feedback, but current on-policy RL runtimes execute rollout, reference scoring, and actor training in strict serial phases. While effective for text-only RL, this phase-granular execution is wasteful for VLMs, where processing dense video inputs and prompt prefixes occupies a large fraction of each phase. Because prefix processing is independent of the generated response, it can be run alongside rollout decoding, which leaves GPU compute capacity underutilized, without breaking synchronous on-policy semantics. We present Rollplex, a runtime that decomposes the reference and training phase and moves the prefix computation into the rollout decode window. Realizing this schedule requires more than concurrent kernel launches: naive colocation of Qwen2.5-VL-32B requires roughly 165 GiB per GPU, while rollout and training prefer different tensor-parallel (TP) degrees and weight layouts. Rollplex addresses these constraints with two mechanisms. Phase-aware memory management controls HBM residency according to producer-consumer lifetimes. Parallelism-aware weight sharing uses the same physical storage for layout-compatible tensors across distinct TP degrees and reconstructs only incompatible tensors, avoiding a complete second actor copy. On 32 H800 GPUs, Rollplex achieves $1.23\times$ – $1.30\times$ speedup over serial colocation and $1.57\times$ – $2.24\times$ over disaggregation under the same GPU budget, while preserving the synchronous RL update.

Chinese Summary

研究背景与问题

视觉语言模型（VLM）赋予具身智能体从视觉观察和语言指令中进行推理与行动的能力。强化学习（RL）后训练通过任务反馈进一步提升这些能力，但当前的同策略（on-policy）RL运行时采用**严格的串行阶段**执行：rollout（采样）、reference scoring（参考评分）和actor training（策略训练）依次进行。这种阶段粒度执行在纯文本RL中有效，但对VLM而言极为低效——处理密集视频输入和提示前缀（prefix）占据了每个阶段的大量计算时间。

核心洞察

前缀处理与生成响应**相互独立**，因此可以将其与rollout解码并行执行，而不会破坏同策略同步语义。然而，当前阶段串行执行导致GPU计算能力严重闲置。

方法创新：Rollplex

Rollplex提出将reference和training阶段进行分解，把前缀计算**迁移至rollout解码窗口内**并行执行。实现这一调度面临两大挑战：

- **显存压力**：Qwen2.5-VL-32B的朴素共置需要约165 GiB/GPU，远超单卡容量
- **并行度冲突**：rollout与training偏好不同的张量并行（TP）度和权重布局

Rollplex通过两种关键机制解决上述问题：

1. **阶段感知显存管理**：依据生产者-消费者生命周期控制HBM驻留，动态释放不再需要的中间数据
2. **并行度感知权重共享**：对不同TP度下布局兼容的张量复用同一物理存储，仅重建不兼容张量，避免维护完整的第二份actor副本

主要结果

在32张H800

GPU上，Rollplex相比串行共置获得** $1.23\times-1.30\times$ 加速**，相比同GPU预算下的分离部署获得** $1.57\times-2.24\times$ 加速**，同时完全保持同步RL更新语义。

意义

Rollplex首次证明：在不改变RL算法语义的前提下，通过跨阶段GPU空间共享和智能显存管理，可显著提升VLM后训练效率，为大规模多模态RL训练提供了新的系统优化方向。

English Summary

Core Contribution

Rollplex is a runtime system that enables **cross-phase GPU spatial sharing** for vision-language model (VLM) post-training. It identifies that prefix computation—which processes dense video inputs and prompt tokens—is independent of response generation, yet current on-policy RL runtimes execute rollout, reference scoring, and training in strict serial phases, leaving GPU compute underutilized.

Key Mechanisms

Rollplex decomposes reference/training phases and overlaps prefix computation with rollout decoding. To handle the 165 GiB/GPU memory requirement of naive colocation, it introduces: (1) **phase-aware memory management** that controls HBM residency via producer-consumer lifetimes, and (2) **parallelism-aware weight sharing** that reuses physical storage for layout-compatible tensors across different tensor-parallel degrees, avoiding a full second actor copy.

Results

On 32 H800 GPUs, Rollplex achieves ** $1.23\times-1.30\times$ speedup** over serial colocation and ** $1.57\times-2.24\times$ speedup** over disaggregation under the same GPU budget, while preserving synchronous RL update semantics. This demonstrates that significant efficiency gains are possible without altering RL algorithm behavior, offering a practical system-level optimization for large-scale multimodal RL training.

Paper 14

Generating Benchmark Health Data Using a Tabular Diffusion Transformer

Authors: Hao Yan, Lisa Pilgram, Dan Liu, Linglong Kong, Fida Dankar et al.

Categories: cs.LG, cs.AI

URL: <https://arxiv.org/abs/2608.14496v1>

Keywords: model, transformer, diffusion

Source

Abstract

Cross-Tabular Data Generation (CTDG) seeks to learn a generative model from multiple heterogeneous tables and produce new synthetic tabular datasets. However, existing synthetic tabular data generation methods are largely restricted to single-input-table scenarios and struggle to effectively handle multiple heterogeneous tables with diverse feature sets. To address this limitation, we propose a two-stage framework for cross-tabular data generation. In the first stage, each heterogeneous raw table is transformed into a standardized statistical table with the same set of columns across all tables.

Each statistical table captures the marginal distributions of the original columns and the pairwise correlations among them. In the second stage, a diffusion transformer model is trained to capture structural patterns across these homogeneous statistical tables and to generate synthetic statistical tables. Synthetic raw tables are subsequently reconstructed from the generated statistical tables via multivariate Gaussian sampling followed by an inverse probability integral transform. This two-stage CTDG framework enables the learning of a unified generative model from multiple heterogeneous tables and supports the generation of an unlimited number of realistic synthetic heterogeneous tables. Experimental results demonstrate high fidelity in the learned statistical representations and a favorable fidelity-diversity trade-off in the generated synthetic data, validating the effectiveness of the proposed approach.

Chinese Summary

研究背景

跨表格数据生成旨在从多个异构数据表中学习生成模型，并产出新的合成表格数据。然而，现有合成表格数据生成方法大多局限于单一输入表场景，难以有效处理具有不同特征集的多个异构表格。

研究方法

本文提出一种**两阶段跨表格数据生成框架**：

- **第一阶段**：将每个异构原始表转换为具有统一列结构的标准化统计表，捕获原始列的边际分布及两两相关性。

-

第二阶段：训练一个**扩散Transformer模型**，学习这些同质统计表中的结构模式，并生成合成统计表。随后通过多元高斯采样和逆概率积分变换，从生成的统计表重建合成原始表。

主要发现与创新

该框架的核心创新在于：

1. 将异构表格统一为同质统计表示，使模型能够从多表联合学习
2. 支持生成**无限数量**的真实感合成异构表格
3. 实验证明，该方法在统计表示保真度上表现优异，并在合成数据的**保真度-多样性权衡**上取得良好平衡

结论

该两阶段框架有效解决了跨表格数据生成的挑战，为多源异构表格的合成提供了可行方案，验证了扩散Transformer在表格生成领域的强大潜力。

English Summary

Core Contribution

This paper proposes a **two-stage cross-tabular data generation (CTDG) framework** that overcomes the single-table limitation of existing methods. The first stage converts heterogeneous raw tables into standardized statistical tables with uniform columns, capturing marginal distributions and pairwise correlations. The second stage trains a **tabular diffusion transformer** to learn structural patterns across these homogeneous statistical tables and generate synthetic ones, from which raw tables are reconstructed via Gaussian sampling and inverse probability integral transform.

Key Results

The framework enables learning a unified generative model from multiple heterogeneous tables and supports unlimited synthetic table generation. Experiments demonstrate **high statistical fidelity** and a **favorable fidelity-diversity trade-off**, validating the effectiveness of this approach for benchmark health data generation and broader CTDG applications.

Paper 15

LP-NAS: Linear Programming-based Neural Architecture Search

Authors: Abhishek Shukla, Ankur Sinha, Faiz Hamid

Categories: cs.LG, cs.AI

URL: <https://arxiv.org/abs/2608.14472v1>

Keywords: neural, algorithm, optimization, design, network

Abstract

Neural Architecture Search (NAS) aims to automate neural network architecture design, reducing reliance on human expertise. Among the various NAS methods, differentiable NAS has gained prominence due to its efficiency and accuracy compared to conventional NAS approaches. Since differentiable NAS relaxes the architecture search space into a continuous domain, it is possible to apply principles from continuous optimization to NAS. In this paper, we propose Linear Programming-based NAS (LP-NAS), a mathematical programming-based framework for differentiable NAS that is applicable to a wide range of continuous search spaces. LP-NAS formulates a linear program (LP) using the validation-loss gradient and the training-loss Hessian to compute an architecture update direction that improves generalization while preserving the optimality of the model parameters. By following this LP-derived descent direction, LP-NAS efficiently navigates the architecture search space, leading to faster and more effective architecture optimization. We introduce two computationally efficient variants of LP-NAS, namely S-LP-NAS and R-LP-NAS. Applying LP-NAS to the Differentiable Architecture Search (DARTS) search space results in two algorithmic variants, S-LP-DARTS and R-LP-DARTS. Both variants achieve faster convergence and significantly higher validation performance during the early search iterations than the standard DARTS algorithm. Extensive experiments on CIFAR-10 and CIFAR-100 show that LP-DARTS outperforms standard DARTS in both the architecture search and evaluation phases. Additionally, we compare our approach with several DARTS variants (P-DARTS, PC-DARTS, and STO-DARTS) on the CIFAR-10 dataset and demonstrate its effectiveness. Furthermore, we validate the transferability of the discovered architectures through experiments on the ImageNet dataset.

Chinese Summary

研究背景与问题

神经架构搜索（NAS）旨在自动化神经网络设计，减少人工依赖。可微分NAS因其高效性和准确性而备受关注，其将搜索空间松弛为连续域，从而可应用连续优化原理。然而，现有方法在搜索早期常面临收敛慢、泛化不足的问题。

方法创新

本文提出**LP-NAS**——一种基于线性规划的NAS框架，适用于广泛的连续搜索空间。核心思想是：利用验证损失梯度与训练损失Hessian矩阵构建线性规划（LP），计算既能提升泛化能力、又保持模型参数最优性的架构更新方向。通过沿该LP导出的下降方向迭代，LP-NAS能高效导航搜索空间，加速架构优化。

我们设计了两种计算高效变体：**S-LP-NAS**（简化版）与**R-LP-NAS**（稳健版）。将其应用于DARTS搜索空间，得到S-LP-DARTS和R-LP-DARTS两个算法变体。

主要发现与结果

- **收敛速度**：两种变体在搜索早期迭代中均比标准DARTS收敛更快，验证性能显著更高。
- **性能提升**：在CIFAR-10和CIFAR-100上，LP-DARTS在搜索与评估阶段均优于标准DARTS。
- **对比实验**：与P-DARTS、PC-DARTS、STO-DARTS比较，在CIFAR-10上验证了其有效性。
- **迁移性**：在ImageNet上的实验证实了所发现架构的良好迁移能力。

意义

LP-NAS将数学规划与NAS有机结合，提供了一种理论清晰、计算高效的架构搜索新范式，尤其适合早期快速探索与资源受限场景。

English Summary

Core Contribution

We propose **LP-NAS**, a mathematical programming framework for differentiable neural architecture search. It formulates a linear program using validation-loss gradients and training-loss Hessian to derive architecture update directions that improve generalization while preserving parameter optimality. Two efficient variants, S-LP-NAS and R-LP-NAS, are introduced and instantiated as S-LP-DARTS and R-LP-DARTS in the DARTS search space.

Key Results

Both variants achieve faster convergence and significantly higher validation performance in early search iterations than standard DARTS. On CIFAR-10/100, LP-DARTS outperforms standard DARTS in both search and evaluation phases. Comparisons with P-DARTS, PC-DARTS, and STO-DARTS on CIFAR-10 confirm its effectiveness, and ImageNet experiments demonstrate the transferability of discovered architectures.

Impact

LP-NAS bridges linear programming and NAS, offering a theoretically grounded, computationally efficient approach for rapid architecture exploration, particularly beneficial in early-stage optimization and resource-constrained settings.

Paper 16

Designing Compact Neural Architectures via Neuron Gating and Mixed Activation

Authors: Abhishek Shukla, Ankur Sinha, Faiz Hamid

Categories: cs.LG, cs.AI

URL: <https://arxiv.org/abs/2608.14443v1>

Keywords: neural, network

Source

Abstract

Neural Architecture Search (NAS) is naturally formulated as a bilevel optimization problem, where the upper-level optimizes the architecture using validation performance and the lower-level trains network parameters using training loss. However, NAS is computationally expensive due to discrete architectural decisions, exponentially growing search spaces, and the high cost of training candidate architectures. This work develops a general bilevel optimization framework for NAS across diverse architectures, including MLPs, CNNs, RNNs, and Transformers, to identify compact architectures with strong predictive performance. We propose three scalable formulations that replace discrete neuron- and activation-level decisions with continuous relaxations, enabling differentiable optimization over otherwise combinatorial architecture spaces. These formulations give rise to three NAS methods: NAS based on Neuron Gating (NAS-NG), NAS based on Mixed Activation (NAS-MA), and NAS based on Neuron Gating and Mixed Activation (NAS-NGMA). Experiments on MLPs and CNNs using MNIST and CIFAR-10 show that the proposed methods consistently identify compact architectures with competitive or improved predictive performance. On MNIST, NAS-NGMA achieves 98.68% test accuracy with 7.69M MLP parameters, while NAS-NG achieves 99.63% accuracy with only 0.26M CNN parameters. On CIFAR-10, the proposed methods consistently outperform vanilla DARTS. Further experiments demonstrate that NAS-NG can optimize substantially over-parameterized and literature-optimal architectures, improving accuracy while reducing parameters. These results establish relaxed bilevel optimization as a scalable alternative to

discrete NAS and provide a general framework for efficient neuron- and activation-level architecture optimization.

Chinese Summary

****研究背景****：神经架构搜索（NAS）通常被建模为双层优化问题——上层基于验证集性能优化架构，下层基于训练损失更新网络参数。然而，离散的架构决策、指数级增长的搜索空间以及候选架构的高训练成本，使得NAS计算开销巨大。

****方法创新****：本文提出一个通用的双层优化框架，适用于MLP、CNN、RNN和Transformer等多种架构，旨在发现兼具紧凑性与强预测性能的架构。我们设计了三种可扩展的松弛公式，将离散的神经元级和激活级选择替换为连续松弛，从而将组合架构空间转化为可微优化问题。由此衍生三种NAS方法：基于神经元门控的NAS（NAS-NG）、基于混合激活的NAS（NAS-MA）以及两者结合的NAS-NGMA。

****实验结果****：在MNIST和CIFAR-10上对MLP和CNN的验证表明，所提方法能稳定发现紧凑且性能优异或更优的架构。在MNIST上，NAS-NGMA以7.69M MLP参数达到98.68%测试精度，而NAS-NG仅用0.26M CNN参数即获99.63%精度。在CIFAR-10上，所有方法均优于经典DARTS。进一步实验显示，NAS-NG能优化高度过参数化及文献最优架构，在减少参数的同时提升精度。

****结论****：该工作确立了松弛双层优化作为离散NAS的可扩展替代方案，并为神经元级与激活级的通用高效架构优化提供了统一框架。

English Summary

This paper proposes a general bilevel optimization framework for neural architecture search (NAS) across MLPs, CNNs, RNNs, and Transformers, targeting compact yet high-performance architectures. Three continuous relaxation formulations—neuron gating (NAS-NG), mixed activation (NAS-MA), and their combination (NAS-NGMA)—replace discrete architectural decisions with differentiable surrogates, enabling scalable optimization over combinatorial spaces. On MNIST and CIFAR-10, the methods consistently find compact architectures with competitive or improved accuracy: e.g., NAS-NGMA achieves 98.68% (MLP) and NAS-NG reaches 99.63% (CNN) on MNIST, while all outperform vanilla DARTS on CIFAR-10. Notably, NAS-NG can also refine over-parameterized or literature-optimal architectures, reducing parameters while boosting accuracy. This establishes relaxed bilevel optimization as a scalable alternative to discrete NAS and offers a general tool for neuron- and activation-level architecture search.

Paper 17

Style or Signature? Artist-Disjoint Evaluation of Style Classification in Frozen Vision Embeddings

Authors: Rory Ashton

Categories: cs.CV, cs.LG

URL: <https://arxiv.org/abs/2608.14435v1>

Keywords: visual, recognition

Source

Abstract

Frozen image embeddings from models such as CLIP are increasingly used to classify paintings by art-historical style, with high reported accuracy. We ask whether this accuracy reflects an understanding of style or the recognition of individual artists. Standard evaluation uses random splits in which works by the same artist appear on both sides, so a classifier can succeed by recognising the painter rather than the movement. We re-evaluate style classification under an artist-disjoint protocol, holding

out every artist in turn so that no work is ever classified using other works by its own painter. On a balanced dataset of 320 paintings across four twentieth-century movements, 5-NN style accuracy falls from 0.87 to 0.77 under this protocol, and the drop is sharply uneven. Impressionism and Cubism barely move, while Surrealism falls twenty points. The pattern holds across four image encoders, including a vision-only self-supervised model, which places the effect in visual structure rather than language. Where an encoder captures genuine shared form, individual artists are barely recognisable yet style is robust, while Surrealism shows the opposite. We argue that artist-disjoint evaluation is necessary to measure stylistic understanding in frozen embeddings.

Chinese Summary

研究背景与问题

CLIP等模型的冻结图像嵌入（frozen

embeddings）被广泛用于绘画的艺术风格分类，并报告了高准确率。本文质疑：这种高准确率究竟反映了模型对**风格**的理解，还是对**艺术家身份**的识别？标准评估采用随机划分，同一艺术家的作品会同时出现在训练集和测试集中，分类器只需识别画家即可“作弊”成功。

方法与协议

我们提出**艺术家不相交（artist-disjoint）**评估协议：逐一留出每位艺术家的全部作品，确保任何测试作品都不会与同画家的其他作品共同参与分类。在包含20世纪四个艺术运动（印象派、立体主义、超现实主义、表现主义）的320幅平衡数据集上，采用5近邻（5-NN）分类器。

主要发现

- 在艺术家不相交协议下，风格分类准确率从0.87**降至0.77**，且下降幅度极不均衡。
- **印象派与立体主义几乎不受影响**（下降<2个点），而**超现实主义下降20个百分点**。
- 该模式在四种图像编码器（含纯视觉自监督模型）中一致复现，表明效应源于**视觉结构**而非语言标签。
-

关键洞察：当编码器捕捉到真正的共享形式特征时，艺术家个体几乎不可区分，但风格分类依然稳健；超现实主义则相反（艺术家易识别，风格难分类）。

结论与意义

艺术家不相交评估是衡量冻结嵌入中**风格理解**的必要条件。现有随机划分高估了模型对风格概念的泛化能力，混淆了“认画家”与“懂风格”。本研究为视觉表征评估提供了更严谨的基准，并揭示了不同艺术运动在嵌入空间中的结构差异。

English Summary

Core Contribution

We demonstrate that high style-classification accuracy in frozen vision embeddings (e.g., CLIP) largely reflects **artist recognition**, not genuine style understanding. Standard random splits leak artist identity across train/test sets, enabling classifiers to shortcut by identifying painters.

Method & Key Results

We introduce an **artist-disjoint evaluation protocol** that holds out all works by each artist. On 320 paintings from four 20th-century movements, 5-NN style accuracy drops from 0.87 to 0.77. The decline is sharply uneven: Impressionism and Cubism remain stable, while Surrealism falls by 20 points. This pattern persists across four encoders, including a vision-only self-supervised model, confirming the effect is visual, not linguistic.

Conclusion

Artist-disjoint evaluation is **necessary** to measure true stylistic generalization. Our findings expose a fundamental limitation

of current embedding benchmarks and highlight that some movements (e.g., Surrealism) lack robust shared visual form, making them inherently harder to classify by style alone.

Paper 18

Designing Reinforcement Learning for Diffusion Models: A Unified Path-Space View

Authors: Yixian Xu, Yuanrui Zhang, Shengjie Luo, Liwei Wang, Di He

Categories: cs.LG, cs.CV, stat.ML

URL: <https://arxiv.org/abs/2608.14430v1>

Keywords: learning, reinforcement

Abstract

Reinforcement learning (RL) post-training provides a direct way to align diffusion models with human preferences and task-specific rewards. However, current RL algorithms for diffusion models remain fragmented: reverse-trajectory methods rely on discretized likelihood ratios, whereas forward-matching methods train on reward-labeled noising versions of the rollout samples. This paper shows that these seemingly different losses arise from a single path-space principle. Starting from the regularized diffusion-RL objective, we use importance sampling between sampling SDEs to obtain an explicit policy-gradient estimator on trajectory space. The estimator contains the stochastic Itô integral underlying Flow-GRPO-type updates; we derive an equivalent variance-reduced value-gradient form that recovers the forward-matching structure of AWM and DiffusionNFT. This identifies the empirical gap between these method families as a variance-reduction effect rather than a difference in RL principle. The derivation yields a unified design space organized by value-gradient estimation, weight functions, and sampling choices. Within this space, we propose a multi-sample KDE value-gradient estimator that reuses rollout groups, together with scale-bounded weight families that retain stable existing recipes while excluding singular ones. Experiments on SD3.5-M and Qwen-Image models validate the variance-reduction explanation and show that the resulting recipe improves over prior diffusion-RL baselines.

Chinese Summary

研究概述

本文针对扩散模型的强化学习（RL）后训练算法碎片化问题，提出了统一的**路径空间（Path-Space）理论框架**，揭示了现有两类主流方法——反向轨迹法与正向匹配法——在数学本质上的内在联系。

核心方法

作者从正则化扩散-RL目标出发，通过采样SDE间的**重要性采样**技术，推导出轨迹空间上的显式策略梯度估计器。该估计器包含Flow-GRPO型更新所需的Itô随机积分；进一步推导出等价的**方差缩减值梯度形式**，恰好还原了AWM和DiffusionNFT的正向匹配结构。这一推导证明：两类方法的经验差异本质上是**方差缩减效应**，而非RL原理不同。

设计空间与创新

基于该理论，论文构建了由**值梯度估计、权重函数、采样选择**三个维度组成的设计空间。在此空间内提出：

- **多样本KDE值梯度估计器**：复用rollout分组，提升样本效率
- **有界缩放权重族**：保留现有稳定配方，排除奇异权重配置

实验结果

在SD3.5-M和Qwen-Image模型上的实验验证了方差缩减解释的有效性，所提方案在多项扩散RL基线中取得一致性能提升。

English Summary

This paper unifies fragmented RL post-training methods for diffusion models under a single **path-space principle**. By applying importance sampling between sampling SDEs to the regularized diffusion-RL objective, the authors derive an explicit policy-gradient estimator that bridges reverse-trajectory methods (e.g., Flow-GRPO) and forward-matching methods (e.g., AWM, DiffusionNFT), revealing their empirical gap as a **variance-reduction effect** rather than a conceptual difference. The derivation yields a unified design space organized by value-gradient estimation, weight functions, and sampling choices, within which they propose a multi-sample KDE value-gradient estimator and scale-bounded weight families. Experiments on SD3.5-M and Qwen-Image validate the variance-reduction explanation and demonstrate consistent improvements over prior diffusion-RL baselines.

Paper 19

CytoBERT: A Foundation Model for Cytometry Data

Authors: Syed Abdul Haseeb Qadri, Bjarne C. Hiller, Felix Blanke, Vanja Sophie Cangalovic, Kutalm Cokun et al.

Source

Categories: cs.LG

URL: <https://arxiv.org/abs/2608.14414v1>

Keywords: foundation, learning, machine, model

Abstract

Cytometry measures the complex characteristics of single cells (e.g., counts and protein expression of immune cells) and is widely used across immunological research and clinical settings. However, cytometry data is highly heterogeneous and unstandardized due to experimental protocols and the choice of measured features. While machine learning methods hold the potential to gain deeper insights into cell biology, these challenges make them difficult to apply and transfer across studies. Recent advances in foundation models can alleviate these issues, but corresponding approaches are still scarce in this field. To address this, we provide CytoBERT, a publicly available, open-source, open-weight foundation model for single-cell cytometry data with variable marker panels. CytoBERT is pretrained in a self-supervised manner on a large-scale cytometry corpus (15 human datasets with heterogeneous marker panels and more than 50 million cells) curated through marker standardization, enabling it to learn transferable inter-marker relationships within cells. Fine-tuning CytoBERT for sample-level classification demonstrates that transfer learning across heterogeneous cytometry datasets is feasible, providing a starting point for scalable, generalizable cytometry analysis. Code is available at GitHub.

Chinese Summary

研究背景

流式细胞术可测量单细胞的复杂特征（如免疫细胞数量与蛋白表达），在免疫学研究和临床诊断中应用广泛。然而，由于实验方案和检测标记物选择的不同，细胞术数据高度异质且缺乏统一标准，这限制了机器学习方法跨研究的应用与迁移。

方法创新

本文提出 **CytoBERT**——首个面向单细胞细胞术数据的开源基础模型，支持可变标记物组合。其核心创新包括：

- **大规模预训练**：在包含15个人类数据集、超5000万细胞的语料库上进行自监督学习，通过标记物标准化统一异质数据
- **可迁移表征**：模型学习细胞内部标记物间的通用关联模式，而非依赖特定实验方案
- **灵活适配**：支持不同实验室、不同标记物面板的数据输入

主要发现

通过微调CytoBERT进行样本级分类任务，实验证明：

跨异质细胞术数据集的迁移学习切实可行

预训练表征可有效泛化至未见过的实验条件

为大规模、可推广的细胞术分析提供了统一基础

开源贡献

模型权重、代码及数据预处理流程已全部公开于GitHub，便于领域研究者直接使用与二次开发。

English Summary

Core Contribution

We introduce **CytoBERT**, the first open-source foundation model for single-cell cytometry data with variable marker panels. Pretrained on a curated corpus of 15 heterogeneous human datasets (50M+ cells) via self-supervised learning, it captures transferable inter-marker relationships within cells.

Key Results

Fine-tuning CytoBERT for sample-level classification demonstrates that cross-dataset transfer learning is feasible despite experimental heterogeneity. The model achieves robust generalization across diverse marker panels, providing a scalable and reproducible baseline for cytometry analysis. All code and weights are publicly available on GitHub.

Paper 20

Offline Deep Q* Estimation with Diffusion Models

Authors: Xiaohong Chen, Yuling Jiao, Lican Kang, Jerry Zhijian Yang, Chen Zhong

Categories: stat.ML, cs.LG

URL: <https://arxiv.org/abs/2608.14401v1>

Keywords: deep, neural, network, learning

Source

Abstract

In offline RL, estimating the optimal action-value function Q^* can be formulated as solving the optimal Bellman equation based solely on offline observations. A fundamental challenge is that the reward function and transition kernel are unknown, so the optimal Bellman operator is not directly observable from data. To address this issue, we propose a novel framework that decouples operator estimation from value function learning. In this approach, we first formulate conditional diffusion models to estimate the reward law and transition kernel, which induces a data-driven approximation of the optimal Bellman operator. We then plug these estimators into the Bellman equation and obtain a deep estimator of Q^* by minimizing the empirical Bellman residual over a neural network function class. Theoretically, we first establish sharp nonasymptotic convergence rates for learning the optimal Bellman operator through an end-to-end analysis of conditional diffusion estimation in total variation distance. We then establish the oracle value-stage rate $\widetilde{O}(\sqrt{n^{-\frac{2}{d_x+d_a+2\beta}}})$ for the excess Bellman residual risk. Finally, under a concentrability condition, we translate this residual bound into an L^2 convergence rate of $\widetilde{O}(\sqrt{n^{-\frac{2}{d_x+d_a+2\beta}}})$ for the resulting deep estimator of Q^* , where d_x and d_a denote the dimensions of the state and action spaces, respectively, and β denotes the Hölder smoothness index of Q^* . Importantly, our theoretical analysis does not rely on completeness assumptions commonly used in deep RL theory. Extensive numerical experiments demonstrate the effectiveness of the proposed method and its strong empirical performance.

Chinese Summary

离线深度Q*估计与扩散模型

****研究背景与挑战****：在离线强化学习中，仅基于静态观测数据估计最优动作价值函数 Q^* ，可转化为求解最优贝尔曼方程。核心难点在于奖励函数与转移核未知，导致最优贝尔曼算子无法直接从数据中观测。

****方法创新****：本文提出一种将算子估计与价值函数学习解耦的新框架。首先，利用条件扩散模型估计奖励分布与转移核，构建数据驱动的最优贝尔曼算子近似；其次，将估计器代入贝尔曼方程，通过最小化神经网络函数类上的经验贝尔曼残差，得到 Q^* 的深度估计器。

****理论贡献****：

- 建立条件扩散估计在总变差距离下的端到端非渐近收敛分析，首次给出最优贝尔曼算子学习的锐利收敛速率
- 推导出超额贝尔曼残差风险的oracle率： $\widetilde{\mathcal{O}}(n^{-\frac{2\beta}{d_x+d_a+2\beta}})$
- 在可集中性条件下，将残差界转化为 Q^* 估计器的 L^2 收敛率： $\widetilde{\mathcal{O}}(n^{-\frac{\beta}{d_x+d_a+2\beta}})$ ，其中 d_x 、 d_a 分别为状态、动作维度， β 为 Q^* 的Hölder光滑指数
- 理论分析不依赖深度强化学习中常用的完备性假设，突破了现有理论框架的局限

****实验验证****：大量数值实验表明，该方法在多种基准任务上表现优异，验证了其有效性与强实证性能。该工作为离线强化学习中的价值估计提供了全新的理论视角与实用工具。

English Summary

Offline Deep Q^* Estimation with Diffusion Models

This paper addresses offline reinforcement learning by estimating the optimal action-value function Q^* through solving the Bellman equation with unknown reward and transition dynamics. We propose a novel framework that decouples operator estimation from value learning: conditional diffusion models first estimate the reward law and transition kernel, inducing a data-driven optimal Bellman operator, which is then plugged into a deep Q^* estimator via empirical Bellman residual minimization. Theoretically, we establish sharp nonasymptotic convergence rates for operator learning in total variation distance, derive the oracle residual risk of $\widetilde{\mathcal{O}}(n^{-\frac{2\beta}{d_x+d_a+2\beta}})$, and obtain an L^2 convergence rate of $\widetilde{\mathcal{O}}(n^{-\frac{\beta}{d_x+d_a+2\beta}})$ for the final estimator under a concentrability condition—all without completeness assumptions. Extensive experiments confirm the method's strong empirical performance.
